

Lecture 1- Part 2: Professional Practice in Computing

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Outline

- **Computer Misuse Act**
- **Data Protection laws**
- Copyright
- Confidential Information

Copyright

Copyright - outline

- Introduce copyright and the laws surrounding it
- Discuss copyright with respect to computer software / code
- Summarise alternatives to copyright

**But first let's talk
about theft**

What is theft?

- Theft: the act of taking someone's property with the intention of depriving them of it
- 2 types of theft you need to be familiar with:
 - Tangible theft
 - Intangible theft

Tangible theft

- Stealing a bicycle, stealing a computer, stealing a jacket, etc
- The object can only be possessed by one person
- It is a physical object that can only be taken from one person by another
- Subject to the law of theft

Intangible theft

- A formula, a poem, “Purple Rain” by Prince, etc
- Can be possessed by many people at the same time
- Subject to intellectual property rights

**But how am I
protected by the law
of theft?**

Theft Act 1968

- A person is guilty of theft if he dishonestly appropriates property belonging to another with the intention of **permanently depriving the other of it**; and “thief” and “steal” shall be construed accordingly.
- 4(1) “Property” includes money and all other property, real or personal, including **things in action and other intangible property**.
- 6(1) ...intention of permanently depriving the other of it **if his intention is to treat the thing as his own** to dispose of regardless of the other’s rights...”

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Note: there’s a permanence to the nature of the theft

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Note: it does make provisions for intangible theft

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Note: taking ownership away from one person to treat the thing as your own

**What about the theft
of Software?**

Copyright + Intangible Theft

- This is what we are most interested in with regards to software and code
- What are you allowed to do with an intangible object that you have not created?

1988 Copyright Design and Patents Act

1. Copyright is a property right which subsists of in accordance with this Part in the following descriptions of work—

- a) original literary, dramatic, musical or artistic works,
- b) sound recordings, films or broadcasts
- c) the typographical arrangement of published editions

- Note: software/code falls under (a)

Copyright is a property right

Property right:

The rights given to the person or persons who have a right to own the property through purchase or bequest. These are basic rights in any society (though absolute right for a property is rare in any society).

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- **Section (a) is the part most relevant to us (it covers code/software)**

1988 Copyright Design and Patents Act

3.1. “literary work” means any work, (other than a dramatic or musical work), which is written, spoken or sung, and accordingly includes—

- a) a table or compilation other than a database
- b) a computer program
- c) preparatory design material for a computer program
- d) a database

1988 Copyright Design and Patents Act

3.2. Copyright does not subsist in a literary, dramatic or musical work unless and until it is recorded, in writing or otherwise

- Note: Presenting something orally isn't enough, it's not copyrighted until it has been recorded

1988 Copyright Design and Patents Act

3.3. It is immaterial for the purposes of subsection (2) **whether the work is recorded by or with the permission of the author**; and where it is not recorded by the author, nothing in that subsection affects the question whether copyright subsists in the record as distinct from the work recorded

- Note: The recording can be an audience recording

A note about databases

1. In this Part “database” means a collection of independent works, data or other materials which—

- a) are **arranged in a systematic or methodical way**, and –
- b) are **individually accessible by electronic or other means**.

A note about databases

2. For the purposes of this Part a literary work consisting of a database is original if, and only if, by reason of the **selection or arrangement of the contents of the database the database constitutes the author's own intellectual creation**

Note: a simple sort or standard way of ordering the information doesn't constitute original intellectual creation. There needs to be an additional contribution of originality within the ordering.

1,000,000 entries with an ordering that constitutes an original intellectual creation does justify having a property right to the database

What if I don't want to copyright my work

Open source software: if you don't want copyright

Public domain: relinquishing all copyright

GNU licences: if you use the code, then any code you produce as a result must also be covered by a GNU licence

Other open source licences give away different rights to: making copies, giving copies away, making changes, use within products for commercial gain

So what is Copyright?

“A copyright is a type of intellectual property that gives its owner the exclusive right to copy, distribute, adapt, display, and perform a creative work, usually for a limited time”

A legal term describing the ownership of control of the rights to use and distribute certain works of creative expression

Copyright License and Transfer

If you attempt to use copyright without the permission of the holder then you are in violation of copyright infringement

As a copyright holder you can:

- License the copyright to someone (give them permission to use it for a defined project)
- Transfer the copyright to someone (e.g. sell it or give it to them)

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Copyright License and Transfer

Licensing copyright example:

- Disney owns the copyright to Star Wars
- They licence the copyright to various video game developers (not owned by Disney) to make Star Wars video games

Copyright transfer example:

- George Lucas owned the copyright for Star Wars
- Eventually he sold it to Disney (transferred the copyright to them for a a LOT of money)
- Disney then started making various Star Wars films and series

Copyright ownership can get messy

Copyright law can be a mess

- Multiple parties involved competing for as much money as possible Company acquisitions and bankruptcies can even result in copyright being “lost”

- Uncertainty over who actually owns it

Company A owns the copyright, they merge with B, then they go bankrupt, company C buys part of company AB’s IP, company D buys another part of it, etc

Maybe company C buys the IP in bulk but only care about one thing

Copyright must be enforceable

You can't just copyright everything

It must be of a suitable scope

- You can't copyright the entire market of something

example: You can't have 1 company with the exclusive right to make card games!

Design as a language of best practices which emerge

- You can copyright your implementation, but you can't copyright the idea conceptually as a whole

Defined Scope

you can't monopolise! Conceptually, the idea of books, films, software, music, etc

The difference between copyrighting all games vs your game

You can copyright your implementation of these

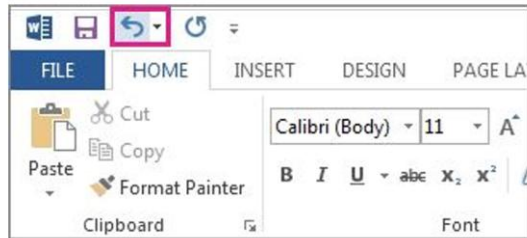
- But not, conceptually, the idea of the medium
- So you protection of your creation but not a monopoly

Design as a language

Design is a language which emerges and evolves over time:

- A gradual process of figuring out best practices
- To avoid reinventing the wheel
- The use of conventions to avoid users needing to re-learn everything
 - New users can quickly learn and understand new systems ----
 - Use of design consistency: Example - video games ----

Example: Undo (can copyright your icon but not the feature)



Left: Google Documents, Right: Microsoft Something

you can copyright your implementation but not the idea

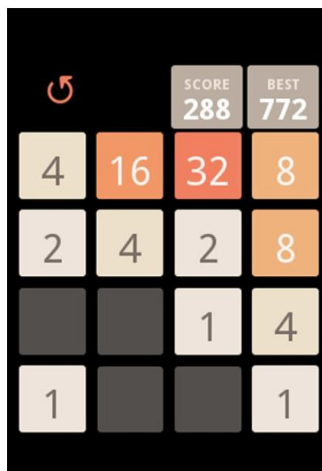
Good for everyone:

Ensures consumer get the best possible experience

Ensures the marketplace is competitive

But it has its shortcomings as well. Example: the story of three!

- Who has played 1024 or 2048 ?



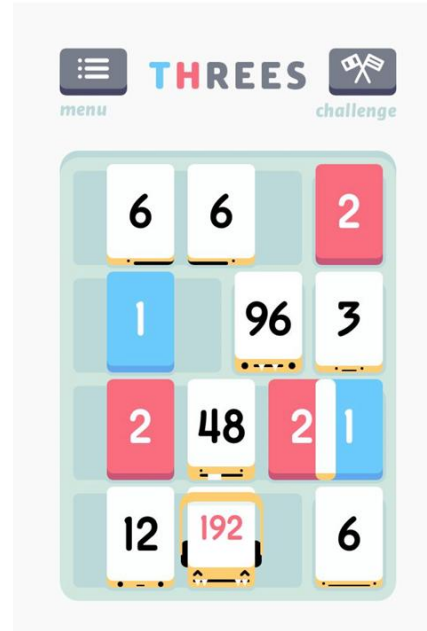
So these are all “clones” of the game Threes!

The creators of Threes! spent 14 months developing their game

-Note: it was a mobile game you had to purchase (currently £4.99)

Just 21 days after Threes! release free clones were available on the app store :1024, 2048, etc – these all quickly overshadowed Threes!

The Threes! developers attempted to get these removed but eventually gave up



Why is this allowed to happen?

- Video games fall into the same area as software copyright
- The underlying source code as well as art and other assets qualify for copyright but the gameplay design itself often does not
- Again back to this idea of something being "beyond the scope of single ownership"

**Confidential
Information**

The law of confidential information

Designed to prevent someone revealing information they have received in confidence

Requires an understanding (an “obligation of confidence”) that the information was given in confidence

It is not required to be written down as explicitly being confidential

The law of confidential information

- A useful tool for protecting commercially sensitive material such as trade secrets which cannot be fully protected by intellectual property rights
- Based around the principle a person who has received information in confidence should not use or disclose the information without permission
- Unauthorised use or disclosure of such information may give rise to an action for breach of confidence

How can information be given in confidence

Imposed by a contract - e.g. NDA (non-disclosure agreements)

Implied by circumstances of discourse - “determined on the basis of whether a reasonable man standing in the shoes of the recipient of the information would have realised that the information was being given to him in confidence”

Implied because of a special relationship between the parties - e.g. employee-employer

A quick note on trade secrets

Trade secrets are a particular type of confidential information

- These are novel techniques used to develop particularly unique products
- An important but invisible part of a company's intellectual property
- Must be protected to stop people just sharing them with their competitors (e.g. disgruntled employees wanting a new job)

Example 1: Howard and Cantor Fitzgerald

- Howard and Harland and two trainees left one financial services firm Cantor Fitzgerald (CF), and joined another one, Tradition
- Developed a similar, but significantly better system than the CF system in three months
- CF claimed that Tradition had infringed their copyright, saying that the Tradition system was a copy of their own.
- It was found that:
 - the CF staff took a copy of the CF code with them to Tradition
 - Harland logged into the CF system after joining Tradition and took a copy of a data file
 - Harland and Howard initially wanted to create a system that would have been a copy of the CF system, and intended to hide the fact that it was a copy
 - the final system was much better, and had less than 4% of CF code
 - Howard had denied any copying, so Tradition spent a lot of money on the defence case
- CF claimed a novel inter-process communication method was a 'trade secret' and that the programmers had breached the obligation of confidence

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Was this a breach of copyright? Was it a breach of confidence?

The Judge's ruling

- The four former CF employees had infringed copyright
- Their employer, Tradition, was also liable, because they were the employers
- Extent of liability (ie. the size of the penalty) was limited because only a little bit of the code had actually been copied
- Howard had to promise to pay Tradition all costs and damages
- With respect to the breach of confidence:
 - The method was a general application programming technique.
 - It would have breached confidence if the programmers had revealed the method while being employed by CF, but it was reasonable to use it as an ex-employee, as it was not a true *trade secret* (simply a useful technique)
 - However, the fact that the programmers used the CF code copy for testing and reference was a breach of confidence

Example 2: John Carmack, Oculus and ZeniMax Media

John Carmack, who previously worked for ZeniMax, and was accused of stealing code and destroying evidence after he started working at Oculus

Found not guilty because it was determined the methods were general programming techniques and he had not copied any code



Summary - The law of confidential information

- The law of confidential information is designed to prevent someone revealing information they have received in confidence
- It does not have to be explicit
- Be aware of what constitutes a breach with respect to code

Lecture Summary

- Computer Misuse Act
- Data Protection laws
- Copyright
- Confidential Information

Video assessed exercise [worth 20%]

- Group exercise (groups of 5 or 4)
- **Groups are already assigned randomly (check Moodle)**

The task : [Create a 90 second video (where the definition of “video” is intentionally vague)]

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